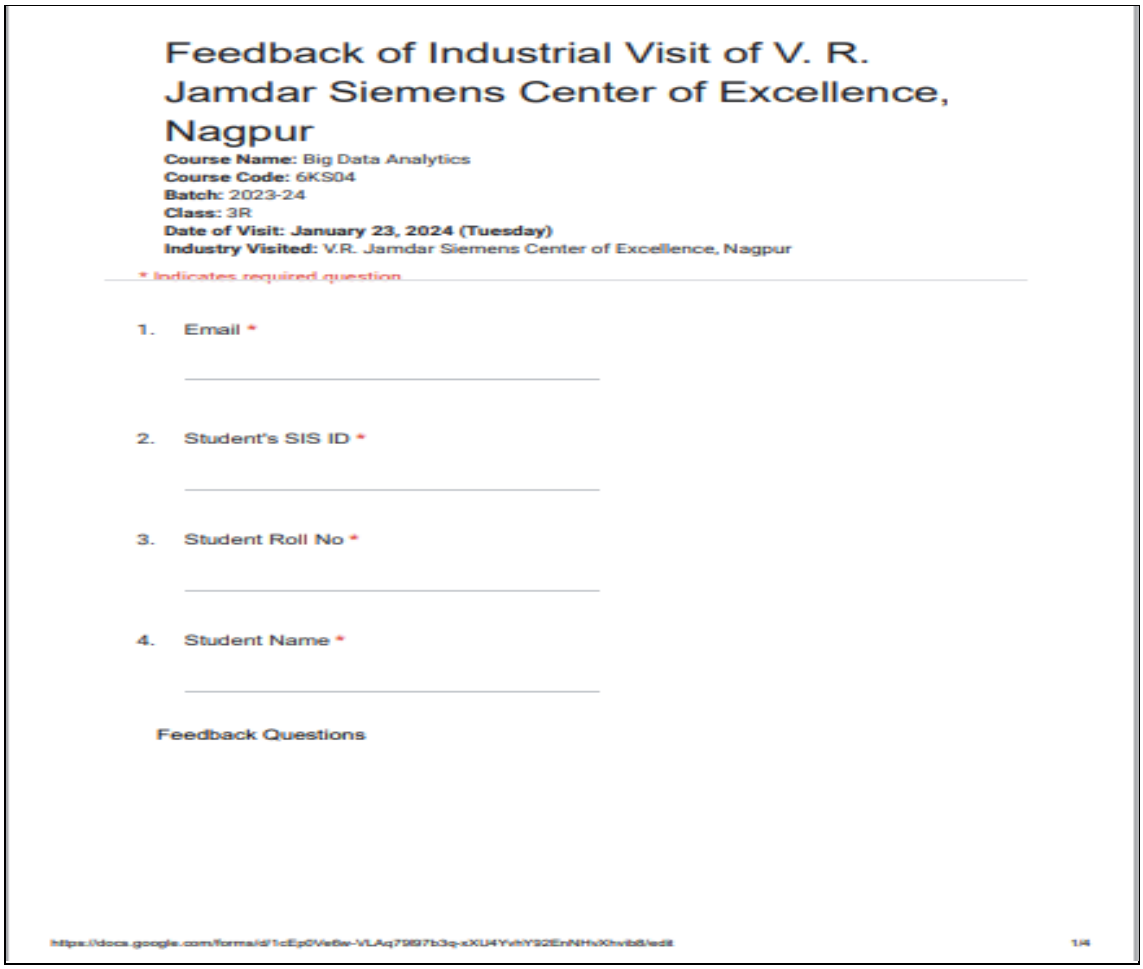
B) The Work Available for Peer Review and Critique

- A **Google Form link** was made available on the college website under the "Innovative Teaching Practices" section.
- Along with the link, a **brief description of the activity**—*Learning through Industrial Visit and Report Writing for Big Data Analytics*—was shared for academic transparency.
- **Academic peers** from the department and other institutions were invited to **review the methodology**, provide **suggestions**, and rate the effectiveness of the learning outcomes.
- The feedback collected was **analyzed thoroughly**, and **constructive inputs** were used to:
    - Enhance the structure of student report submissions.
    - Improve coordination and learning material provided during visits.
    - Incorporate reflective questions in student presentations for deeper learning.
- This peer review process ensured **academic accountability** and promoted a **culture of continuous improvement** in innovative teaching strategies.



Feedback of Industrial Visit of V. R. Jamdar Siemens Center of Excellence, Nagpur

Course Name: Big Data Analytics
Course Code: 6KS04
Batch: 2023-24
Class: 3R
Date of Visit: January 23, 2024 (Tuesday)
Industry Visited: V.R. Jamdar Siemens Center of Excellence, Nagpur

* Indicates required question

1. Email *
2. Student's SIS ID *
3. Student Roll No *
4. Student Name *

Feedback Questions

https://docs.google.com/forms/d/1cEp0Ve6w-VLAq79897b3q-xXU4YvhY92EnNHvXhvibB/edit 1/4

5. 1.How relevant was the field visit to your Big Data Analytics coursework? *

Mark only one oval.

- Very Relevant
- Relevant
- Neutral
- Not Relevant

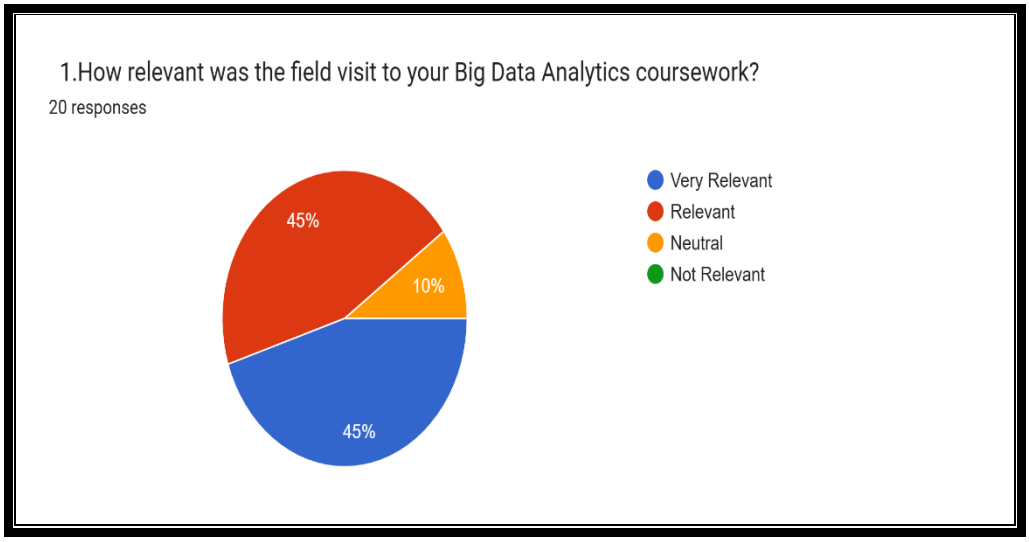
6. 2. What were the key takeaways from the visit regarding Big Data technologies and applications? *

7. 3. Did the visit enhance your understanding of real-world challenges and solutions in Big Data Analytics? *

Mark only one oval.

- Yes
- No
- Maybe

- **Responses samples**



Subject	Teaching Method	Peer Review Feedback	Actions Taken / Planned
Big Data Analytics (Dr. R. A. Zamare)	Learning through Industrial Visit and Report Writing	• Report writing criteria were not well-structured initially. • More technical linkage to Big Data tools (like Hadoop, Spark) was suggested.	✅ Introduced a structured template for report writing. ✅ Included a technical questionnaire covering Hadoop/Spark concepts.

Original Question	Revised Question After Feedback
1.How relevant was the field visit to your Big Data Analytics coursework?	Define Big Data with reference to the 5 Vs (Volume, Velocity, Variety, Veracity, and Value). How were these aspects observed during the industrial visit?
2. What were the key takeaways from the visit regarding Big Data technologies and applications?	Explain the architecture of Hadoop. How was Hadoop implemented in the organization visited, and what specific challenges were highlighted by the experts?
3. Did the visit enhance your understanding of real-world challenges and solutions in Big Data Analytics?	Describe the working of the Hadoop Distributed File System (HDFS). During the field visit, what kind of data was managed using HDFS and what were the storage strategies used?

Original Question	Revised Question After Feedback
4. Were the professionals at Siemens Center of Excellence able to explain the applications of Big Data clearly?	Any specific areas or technologies you would like to explore further at Siemens Center of Excellence ?
5. Which Big Data tools, techniques, or technologies did you observe during the visit?	List applications of Big Data. Identify and elaborate on at least two real-world Big Data applications observed during the field visit, and analyze their impact on decision-making.
6. How effectively did the visit connect theoretical learning with practical implementation?	Explain data ingestion in Big Data systems. During the industrial visit, what tools and techniques were used for data ingestion, and how did they handle data variety and velocity?
7. Did you gain insights into how Siemens uses Big Data in real-world applications?	Compare NoSQL databases with traditional relational databases. How were NoSQL solutions applied in the visited industry, and what were their advantages in handling unstructured data?
8. What new insights did you gain about career opportunities in Big Data Analytics?	Explain the concept and workflow of MapReduce with an example. How was MapReduce utilized in the visited facility, and what were its limitations in their context?

- **Peer Review and Critiques of Innovative Teaching Methods for few subjects is tabulated in table 5.5.1-c**

Subject	Teaching Method	Peer Review Feedback	Actions taken/Planed
Big Data Analytics (Dr. R. A. Zamare)	Learning through Industrial visit/field work and report writing	• Report writing criteria were not well-structured initially. • More technical linkage to Big Data tools (like Hadoop, Spark) was suggested.	☑ Introduced a structured template for report writing. ☑ Included a technical questionnaire covering Hadoop/Spark concepts.

C. The work reproduced and developed further by other scholar:

Title of the work	Faculty Conducting the Original Work	Name of the faculty producing/developing the Work	Actions for Reproduction/Development
Learning through Industrial visit/field work and report writing	Dr. R.A.Zamare	Dr. P.K. Bharne	Categorize questions as **per subject orientation in industrial learning**.

Course Title & Course Code: Big Data Analytics (5KS05)

Class: Third Year CSE Semester: VI

Name of the Course Teacher: Dr. R. A. Zamare

Title of the Innovative Practice: Field Visit to V. R. Jamdar Siemens Center of Excellence VNIT, Nagpur

Objectives / Goals of the Practice

• To provide students with real-world exposure to Big Data Analytics technologies.
• To explore the infrastructure of VNIT's V. R. Jamdar Siemens Center of Excellence.
• To understand practical applications of Industry 4.0 and data-driven systems.
• To promote interdisciplinary innovation through live demonstrations and expert interactions.

Use of Appropriate Methods

• Conducted a full-day industrial field visit to VNIT, Nagpur on 23rd January 2024.
• Students visited 11 advanced laboratories such as IoT Lab, Smart Factory Lab, and Robotics Lab.
• Demonstrations on real-time data analytics using Siemens MindSphere and automation systems.
• Interactive Q&A sessions between students and VNIT lab experts.
• Hands-on exposure to technologies including Digital Twin, IIoT platforms, and AI-integrated manufacturing.

Significance of Results

• Students gained practical knowledge of how big data is processed in smart industries.
• Exposure to real-time analytics, cloud platforms, and predictive maintenance systems.
• Inspired student-led project proposals in domains like IoT Analytics, Smart Manufacturing, and AI-based monitoring.
• Strengthened theoretical understanding by connecting classroom learning to industrial practices.

Effective Implementation

• Field visit was well-coordinated with VNIT's expert team, ensuring guided demonstrations at each lab.
• Labs covered: Reverse Engineering Lab, IoT Lab, Process Instrumentation Lab, Robotics Lab, etc.
• Concepts like Digital Twin, AI + Big Data, Smart Manufacturing were illustrated with working models.
• Faculty facilitated group reflection and encouraged documentation through report writing.

POs Mapped

PO1, PO2, PO3, PO5, PO7, PO12

Reflective Critique

Faculty observed increased curiosity and engagement among students throughout the visit. The hands-on learning environment helped students connect abstract big data concepts with real-world applications. Students suggested including a post-visit brainstorming session to discuss innovative ideas inspired by the visit.

Feedback Link: https://forms.gle/2K4qeCubwbUdC4PY9

Evidences of Success

- Students submitted reflective reports demonstrating clear learning outcomes.
- Increased interest in AI and Big Data-related project proposals post-visit.
- Positive feedback from students on the usefulness of the field experience.
- Engagement in report writing and post-visit discussions showed high academic motivation.

Report Link: 6.0 Quarter 2_VNIT_Field Visit _CSE_Dept.doc